



Superstructure optimization models for regional coal industry development considering water resources constraints—A case study of Ordos, China

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ARTICLE INFO

Keywords:

Structure optimization
Coal industry
Resource constraint
Carbon dioxide emission

ABSTRACT

Optimizing the industrial structure is an important way to address resource constraints, achieve sustainable development. In this study, a planning method that combines the superstructure modeling concept and the mathematical method of mixed integer programming is proposed to investigate the optimized results of the coal industry under given constrained water resources, with 4 major coal application industries and 14 coal technology categories are considered. The results indicate that an increase in available water resources is associated with an increased local utilization rate of coal. Under the Water Rights Replacement Scenario, the local utilization rate of coal reaches 24.9%. Changes in available water resources significantly impact the coal chemical industry. In addition, significant reduction by 25% in carbon dioxide emissions requires 70 percent reduction in IDCL and CTO production. The policy implications for the development of the coal industry are also provided in the end.

1. Introduction

Coal has been crucial to China's economic growth and secure energy supply, and has underpinned the country's rapid economic growth for more than four decades (Liu et al., 2021). According to the China Energy Statistical Yearbook, coal accounted for 56.0% of total energy consumption in 2021; however, this figure is decreasing each year. China is rich in coal compared with oil and natural gas, making the country one of the few coal-based energy producers and consumers in the world (Jie et al., 2021). However, the billions of tons of coal consumed each year produce large amounts of carbon dioxide and pollutants; these contribute to climate change (Zuo et al., 2022). This highlights the importance of reducing the use of fossil energy and encouraging a clean and low-carbon energy transformation. At the same time, to ensure stable economic and societal development, it is critical that China build on its energy and resource endowment base, adhere to the principle of building before breaking, and gradually withdraw traditional energy when safe and reliable alternatives are available. This means that coal will continue to play a leading role in China's energy supply (Yang et al., 2016).

China's coal resources are unevenly distributed (Huang et al., 2017). Ordos, a rare energy rich area in China, has become a primary location

providing a stable supply of national energy, and has made important contributions to the country's rapid and stable economic development. In 2021, Ordos produced 670 million tons of coal, accounting for 16.2 percent of China's coal output. Developing diversified applications of coal (Shi et al., 2021; Li et al., 2023) is an important way to encourage the safe, clean, green, and efficient production, utilization, and consumption of coal to achieve sustainable economic development. It is worth noting that China has an unbalanced distribution of water resources and coal. While it has a lot of coal, the northwest region has moderate to significant water shortages. The advancement of China's industrialization and urbanization has enhanced Ordos's strategic position as a national energy and energy chemical base. However, China's economic growth is slowing, and the elasticity coefficient of energy demand is decreasing. The coal industry in Ordos has become prominent due to a lack of overall planning, disorderly industrial competition created by homogeneous development with neighboring areas, ongoing construction of low-end industries, and overcapacity. Therefore, optimizing the coal industry requires avoiding the energy "Golden Triangle" turning in a "resource curse" based on actual demand.

Water, energy, and emissions are interrelated and accompany the entire energy production and consumption process (Misrol et al., 2021; Wang et al., 2018). Some scholars have studied how water resource constraints and carbon emission constraints influence the development

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Nomenclature*Abbreviations*

c	cost per ton of coal utilization technology (CNY/t)
CC	coking coal
CCCI	coal for coal chemical industry
COKE	coke
CPG	coal for power generation
CMW	coal mining and washing
D	maximum transportation volume
DIC	direct liquefied coal
EC	the maximum environmental capacity of CO ₂ (million tons)
ec	the CO ₂ emission per ton of product of coal utilization technology (t/t)
EN	the maximum environmental capacity of NO _x (million tons)
en	the NO _x emission per ton of product of coal utilization technology (t/t)
ES	the maximum environmental capacity of SO ₂ (million tons)
es	the SO ₂ emission per ton of product of coal utilization technology (t/t)
GANGUE	gangue
HAC	high alumina coal
HOC	high oil coal
M	the total revenue of coal industry
p	price per ton of coal utilization technology (CNY/t)
Q	available water resources in coal industry (million tons)
q	water consumption per ton of product of coal utilization

technology (t/t)

TCD	total coal development (million tons)
WCCI	water for coal chemical industry
WDC	washed Coal
WMW	water for mining and washing
w	output of coal utilization technology (million tons)

Products abbreviations

CA	chlor-alkali
CE	coal export
CFB	circulating fluidized bed power generation
COKING	coking
CTM	coal-to-methanol
CTO	coal-to-olefin
DCL	direct coal liquefaction
HACFB	high alumina coal circulating fluidized bed power generation
HAPCPG	high alumina pulverized coal power generation
IDCL	indirect coal liquefaction
PCPG	pulverized coal power generation
SC	semi-coke
SNG	synthetic natural gas

Subscript

d	types of transportation
i	types of coal utilization technologies
m	types of coal
t	planning year

of power generation technology and economic, there is a lack of research on the overall optimization of the coal industry. Huang et al. (2017) integrated a water resource module into a partially balanced energy environment model, comprehensively considered the energy system and water system, and then evaluated the impact of carbon emission and water constraints on power generation technology. Using Hebei Province, China, as an example, Liu et al. (2020) applied a linear programming model to investigate industrial structure adjustments in the context of water and energy constraints; they then discussed the relationship between water, energy resources and economic development. Liu et al. (2023) studied the impact of the interaction between water consumption and the economy on promoting regional high-quality development. Hassan et al. (2022) analyzed the impact of economic policy on greenhouse gas emissions in the background of natural resources and economic development.

As aforementioned, China's energy consumption is dominated by coal, which is concentrated in some regions, forming many industrial cities that "thrive on coal". Under the background of energy transformation, how to sustain the development of these coal cities has become the main problem that needs to be considered (Zeng et al., 2019). Therefore, many studies have analyzed the sustainable development ability of coal cities from different perspectives. For example, Long et al. (2022) adopted a Slack-Based Measure (SBM) model and included environmental constraints in study variables to reveal the differences in and sources of total factor energy efficiency in 35 coal mining cities. Chen et al. (2021) discussed sustainability evaluation and development path selection in China's coal bases from the perspective of a spatial field. The same situation exists in some countries and regions around the world. The relationship between company performance and sustainability management in Brazil's mining sector (Gomes et al., 2014), as well as the social sustainability of mining communities in northern Europe and northwestern Russia (Suopajarvi et al., 2016), have attracted the attention of some scholars.

Evaluating coal industry situation is an effective way to put forward recommendations for well-rounded coal development (Zhao et al., 2021b). Furthermore, some scholars have conducted evaluation on the influencing factors affecting the development of the coal industry. Such as, appraising the ability of 43 typical coal cities to break through an existing lock-in status using three variables: economy, environment, and sustainable development (Hou et al., 2018). Exploring the development process of typical coal cities from a historical perspective and summarizing lessons and transformations experience in the coal development and utilization process (Zhao et al., 2021c). To assess the impact of policy adjustments on the coal industry, scholars have conducted studies on China's overcapacity reduction policy (Zhang et al., 2022), coal resource tax reform (Xu and Xu, 2021), market and enterprise reform (Li and Hu, 2017).

Regional coal industry planning plays a vital role in economic development. Many scholars have carried out studies on the optimization model of green coal production (Cui et al., 2020), the optimization of coal and new energy industry structure in energy enterprises (Kong et al., 2021). In terms of coal utilization technology, it mainly includes coal-fired power generation (Li et al., 2021) and the coal chemical industry (Xiong and Wei, 2017; Xu et al., 2018), which consume large amounts of water and discharge many pollutants and carbon dioxide (Liu et al., 2021). As such, pollutant emission restrictions and water resource constraints have become key factors restricting its development (Shi et al., 2021), especially in areas lacking water resources. In previous studies on coal utilization technology, scholars analyzed the coal industry's technology development path and its economics in the face of constraints, focusing on the coal chemical industry, coal power industry, or a single technology type (Guo and Xu, 2018; Kong et al., 2019; Jiang et al., 2022).

For example, Zhang et al. (2019) calculated the total carbon dioxide emissions from China's coal chemical industry. Zhao et al. (2021) incorporated water resource shortages and indirect water costs into a

technical and economic evaluation model of modern coal chemical technologies (MCCTs). That study evaluated four technical and economic indexes for coal-to-liquid, synthetic natural gas, coal-to-olefin, and coal-to-ethylene glycol. To investigate the short term (2020), medium term (2030), and long term (2050) layout of the coal chemical industry under restricted carbon dioxide emissions, Huang et al. (2018; 2019) proposed a nonlinear programming method. The research of coal power development under the constraint condition is less than coal chemical industry. Setting greenhouse gas emission reduction and coal consumption as constraints, Ji et al. (2018) adopted an imprecise but robust energy management model to optimize the low-carbon power generation planning of a province in China. They then pointed out that limiting carbon emissions and reducing coal consumption would encourage renewable energy development. Li et al. (2021) studied the water-saving potential of the coal power industry from the perspective of its own water-saving performance.

Many studies have investigated the relationship between water-energy-emissions, and have analyzed the sustainable development capacity of typical coal cities, and have evaluated the factors that affect the development of the coal industry. However, few studies have focused on optimizing the entire coal industry in a specific region under the joint constraints of water resources and carbon emissions. As such, the main objectives of this study are to (i) develop a processing method that combines superstructure modeling and a mixed integer programming (MIP) mathematical method, which enriches the regional coal structure optimization research to some extent; (ii) address this topic, using Ordos, China as a case study, taking into account 4 major coal application industries and 14 coal technology categories; (iii) set a Reference Scenario, a Policy Support Scenario, and Water Rights Replacement Scenario, to compare and analyze coal industry development results under the constraints of available water resources, and examine the impact of carbon dioxide emission constraints on the development of coal industry. The results and conclusions provide a valuable reference for optimizing the industrial structure in coal enrichment areas under water resource and carbon emission constraints.

Given this background, the rest of this paper is organized as follows. Section 2 describes the modeling concept and coal industry development patterns, and analyzes the constraints. Section 3 presents an analysis of water resource constraints. Section 4 presents the development optimization results and analyzes changes in the Ordos coal industry and related industries. Section 5 provides a general conclusion and opportunities for further research.

2. Modeling

2.1. Concept and methods

Developing the Ordos coal industry requires an integrated and optimized layout, addressing special coal types in a targeted way. This section evaluates the effects of Ordos' scientific coal industrial structure, and proposes a reasonable production capacity allocation that ensures the sustainable development and utilization of regional resources, given water resource, environmental, transport channel, and carbon emission constraints. This analysis is vital for coordinating energy, environmental, and economic development.

The optimization analysis and modeling concepts are as follows. According to the principles of technology selection based on coal type and capacity determination based on constraints, a dynamic optimization model of the superstructure is applied to quantitatively evaluate an effective coal industrial structure and capacity scale in Ordos under different constraints. This includes meeting water consumption constraints while maximizing economic benefits. Using a sustainable development strategy and effective development planning for Ordos City, the study lists the specific functions and constraint conditions associated with the target economic benefit. The model sets 2020 as the base year and 2025 as the target analysis year. Fig. 1 shows the main

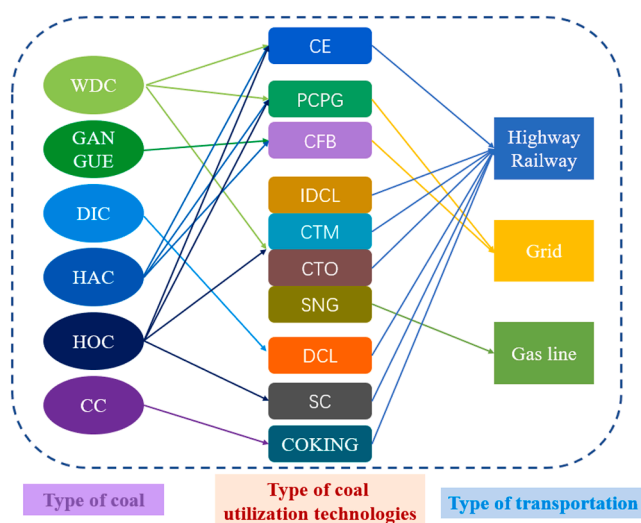


Fig. 1. Coal industry optimization modeling boundary map for Ordos.

factors considered in the optimization research and the basic industrial framework for the optimization analysis.

The optimization analysis includes six types of coal based on Ordos coal resource characteristics; the direction of coal processing, transformation, and utilization; and the feasibility of data acquisition. The six types of coal products include: washing coal (mainly refers to chemical coal and steam coal), gangue (mainly refers to low-quality coal), coking coal, direct liquefaction coal, high-aluminum coal (mainly refers to coal with aluminum oxide making up more than 40% of the coal ash), and high-oil coal (mainly refers to coal with high tar content after raw coal pyrolysis).

Ordos is an important coal producing city in China, about 80% of the coal is transported to other parts of the country for production and life, the rest is utilized locally. The coal industry in Ordos includes power generation, the modern coal chemical industry, and the traditional coal chemical industry. Coal-fired Power generation includes pulverized coal power generation (PCPG, ultra-supercritical units are assumed to be built in the future in the model) and circulating fluidized bed (CFB) power generation (in the model, CFB supercritical units are assumed to be built in the future). Modern coal chemical industry includes direct coal liquefaction (DCL), indirect coal liquefaction (IDCL), coal to methanol (CTM), coal to olefin (CTO), and coal to synthetic natural gas (SNG). Traditional coal chemical industry includes semi-coke (SC), cooking (COOKING), and chlor alkali (CA). The coal used in other industries is uniformly classified as other coal consumption (Others). Low-quality coal and gangue are used locally, but are not exported.

The modeled export transportation modes used to relocate Ordos coal and processed and transformed products include roads, railways, pipelines, and the power grid. The power grid transports pulverized coal power generation and CFB power generation, and the pipeline transports coal-to-SNG. Roads and railways mainly transport direct coal exports and various coal-to-fuel and chemicals.

With respect to the dynamic optimization characteristics shaping the coal industry in Ordos, there are different types of coal and coal utilization technologies available for coal industry development. The existing production capacity should be considered and rationally utilized, as the production capacity of different coal industries evolves from one configuration to another over time.

The superstructure model connects all possible relationships, ensuring that no possible solutions are missed in the solution (Liu et al., 2022). While MIP model is a combination of integer programming, linear programming, 0–1 programming and other planning methods, which can solve large and complex problems involving a variety of decision variables and constraints, and is widely used in production

planning, logistics distribution, route planning and other fields. And MIP model has robustness in dealing with uncertainty and data noise (Leao et al., 2020). The overall idea of this study is as follows. First, the modeling process includes all possible options in each link in Fig. 1 and all possible connections between these options. Then, driven by the optimization objective set, the weak options and connections are discarded through the optimization mathematical algorithm. The options retained and the connections between the options constitute the optimal configuration of each link in constructing the infrastructure (Biegler et al., 1997).

2.2. Mathematical modeling

Fig. 1 shows that each coal utilization technology is composed of coal resources, production technology, and energy carrier transportation. Each link has different options to choose from. As noted above, the goal of capacity optimization of the Ordos coal industry is to maximize economic benefits under the constraint of water consumption.

This section presents a mathematical formula for the study's optimization model, including the objective function and model constraints. Four sets, t , m , I , and d represent time, coal, technology, and transportation, respectively. The physical meanings of the parameters and variables are listed in Nomenclature.

2.2.1. Objective function

The objective function of this model is designed to maximize the benefits of the coal industry, expressed as:

$$\max M = \sum_t \sum_i (p_{it} - c_{it}) w_{it} \quad (1)$$

where M is the total revenue of Ordos coal industry; p_{it} and c_{it} are the price and cost per ton of product, respectively, for the i coal utilization technology in the t planning year; w_{it} is the output of the i coal utilization technology product in the t planning year.

2.2.2. Constraint function

Previous research and analysis indicate that many factors may restrict the future development of the coal industry in Ordos. This section focuses on these factors to analyze the Ordos development setting, including coal resources, transport channels, water resources, SO₂ emissions, NO_x emissions, carbon dioxide emissions, and other variables. The model includes many constraint functions; the most important ones are described and analyzed here.

1) Water consumption constraint function:

The water resource constraint is expressed as:

$$Q \geq \sum_t \sum_i q_{it} w_{it} \quad (2)$$

where Q is the total water available for the Ordos coal industry in 2025; q_{it} is the water consumption of the i coal utilization technology per ton of product in the t planning year; and w_{it} is the output of the i coal utilization technology product in the t planning year.

2) Production constraint function of coal utilization technologies:

The production constraint is expressed as:

$$w_{\min_{it}} \leq w_{it} \leq w_{\max_{it}} \quad (3)$$

where $w_{\min_{it}}$ and $w_{\max_{it}}$ represent the minimum and maximum output, respectively, of the i technical product of coal utilization in the t planning year. The function indicates that the output w_{it} should fall between $w_{\min_{it}}$ and $w_{\max_{it}}$.

3) Resource development constraint function

The resource development constraint is expressed as:

$$DC_{\max_{tm}} \geq \sum_{i \in I_m} w_{it} CC_{it} \quad (4)$$

$$\begin{cases} m = \text{gangue}, i_m = \{\text{CFB}\} \\ m = \text{direct liquefaction coal}, i_m = \{\text{direct liquefaction}\} \\ m = \text{high alumina coal}, i_m = \{\text{pulverized coal power generation, CFB}\} \\ m = \text{high oil coal}, i_m = \{\text{semi-coke}\} \\ m = \text{coking coal}, i_m = \{\text{coking}\} \\ m = \text{washed coal}, i_m = \{\text{other}\} \end{cases}$$

where $DC_{\max_{tm}}$ is the maximum amount of m coal type developed in the t planning year; and CC_{it} is the coal consumption rate of the i coal utilization technology in the t planning year.

4) SO₂ emission constraint function

The SO₂ emission constraint is expressed as:

$$ES \geq \sum_i es_{it} w_{it} \quad (5)$$

where ES is the maximum environmental capacity of sulfur dioxide (SO₂) in Ordos; es_{it} is the SO₂ emissions per ton of product produced by the i coal utilization technology in the t planning year.

5) NO_x emission constraint function

The NO_x emission constraint is expressed as:

$$EN \geq \sum_i en_{it} w_{it} \quad (6)$$

where EN is the maximum environmental capacity of various nitrogen oxides (NO_x); en_{it} is the NO_x emissions per ton of product produced by the i coal utilization technology in the t planning year.

6) Carbon dioxide emission constraint function

The carbon dioxide emission constraint is expressed as:

$$EC \geq \sum_i ec_{it} w_{it} \quad (7)$$

where EC is the maximum environmental capacity of carbon dioxide; ec_{it} is the carbon dioxide emitted from i coal utilization technology per ton of product in the t planning year.

7) Transportation volume constraint function

Transportation of coal-based products in Ordos is done through pipelines, power grids, roads, and railways. The transportation volume constraint is expressed as:

$$DC_{\max_{td}} \geq \sum_{i \in I_d} w_{it} (1 - \varepsilon_{it}) \quad (8)$$

$$\begin{cases} d = \text{pipeline}, i_d = \{\text{SNG from coal}\} \\ d = \text{Power grid}, i_d = \{\text{Pulverized coal power generation, CFB}\} \\ d = \text{highway and railway}, i_d = \{\text{other}\} \end{cases}$$

where $DC_{\max_{td}}$ is the maximum throughput of d transportation mode in the t planning year; and ε_{it} is the internal consumption rate of i coal utilization technology in the t planning year.

2.3. Analysis of development constraints in Ordos

2.3.1. Analysis of the coal resource constraint

The quantities of resources in Ordos at the end of 2015 were as follows: 303.18 billion total tons of coal resources were explored and obtained, including 292.17 billion tons of reserves and 101.69 billion tons of recoverable reserves. Based on the "Energy Comprehensive Development Plan of Ordos during 14th Five-Year", by 2025, the raw coal production capacity should be controlled within 930 million tons, and the output should be controlled within 780 million tons (Ordos Municipal People's Government, 2022). Based on a research investigation, and combined with relevant Ordos planning information, Table 1 shows the maximum exploitable amount of coal resources projected for 2025.

2.3.2. Transport channel constraint analysis

Ordos is an important coal output base in China, and constructing export channels is a key project for developing coal and related

Table 1

List of maximum coal production in Ordos in 2025 (million tons).

	Coal cleaning	Gangue	Direct liquefied coal	High-sulfur coal	High-oil coal	Coking coal	Total coal
Maximum output	625	55	18	32	20	30	780

industries in the region. The Ordos City government has identified several strategic actions, including improving the road network, speeding up the construction of a modern railway network, and promoting the coordinated development of power grids at different voltage levels. Given this, roads, railways, and power grid transmission channels are not expected to be major constraints on the development of the Ordos coal industry.

2.3.3. Environmental constraint analysis

In terms of wastewater discharge, the Shendong mining area of Ordos has utilized mine water resources through an underground reservoir, mine water surface treatment, and other mechanisms. This has turned water waste into valuable materials, and no longer poses constraints to the local environment. As such, the main environmental constraints include gaseous pollutants, such as SO₂ and NO_x.

The requirements of the "Comprehensive Work Plan on Energy Conservation and Emission Reduction of Ordos during 13th Five-Year", state that in 2020, sulfur dioxide and nitrogen oxide emissions in Ordos should be controlled within 184,200 tons and 173,500 tons, respectively. The "Implementation Plan of the 14th Five-Year Plan for Energy Conservation and Emission Reduction in Inner Mongolia Autonomous Region" requires that the cumulative pollutants during the 14th Five-Year period be reduced by more than 10%. Therefore, sulfur dioxide and nitrogen oxides emissions in Ordos should be controlled within 165,800 tons and 156,100 tons, respectively, by 2025.

2.4. Coal related industry development status and technical and economic analysis

2.4.1. Development status of coal related industries in Ordos

Ordos mainly supports coal-to-oil and gas industries, including direct coal liquefaction, indirect coal liquefaction, coal-to-natural gas, and ethylene glycol production. Table 2 indicates that these production capacities are expected to continue to operate in 2025.

The total installed thermal power capacity of Ordos reached 22.99 gigawatts (GW) in 2020. The Energy Comprehensive Development Plan of Ordos during the 14th Five Year states that the installed power generation capacity is expected to reach 97.66 GW in 2025, including 50 GW of renewable energy and 47 GW of thermal power.

2.4.2. Analysis of coal utilization technology

Water consumption is an important parameter for different coal utilization technologies. The water consumption and carbon dioxide emissions per unit product in China are shown in Table 3, it assumes that the values do not change in 2025.

3. Water resource constraint analysis

Ordos is located in the northwest arid area of China, and has more serious water shortages than other regions in China. The average annual

Table 2

Ordos modern coal chemical production projects.

Route	Unit	Capacity
Direct coal liquefaction	million tons/year	1.08
Indirect coal liquefaction	million tons/year	0.16
Coal to SNG	hundred million cubic meters/year	4.0
Coal to olefin	million tons/year	2.60
Coal to methanol	million tons/year	5.90

Table 3

Technical characteristics of coal related utilization.

Techniques	Water consumption	Carbon dioxide emissions	Unit	Ref.
Coal mining and washing	0.20	/	t/t	(Ordos Municipal People's Government, 2017)
Pulverized coal power generation	3.10	4.16	t/ kW	(Chinese Academy Of Engineering, 2017)
CFB Power generation	2.67	4.43	t/kW	(Chinese Academy Of Engineering, 2017)
Indirect coal liquefaction	11.77	8.64	t/t	(Chinese Academy Of Engineering, 2017; Zhang et al., 2019)
Coal to methanol	8.38	2.65	t/t	(Zhang et al., 2019)
Coal to olefin	14.50	9.34	t/t	(Jianchao et al., 2022)
Coal to SNG	6.28	5.15	t/ kilostere	(Chinese Academy Of Engineering, 2017)
Direct coal liquefaction	8.03	3.48	t/t	(Xie, 2014)
Semi-coke	0.50	0.58	t/t	(Xie, 2014; Zhang et al., 2019)
Coking	1.48	0.15	t/t	

precipitation is 190–300 mm, with an average annual evaporation rate of 2000–3000 mm. The per capita water resources are 1318 m³, which is significantly below the national average of 2099 m³. Developing and utilizing coal requires a large amount of water, and water is an important constraint for coal and its related industries. In this section, a scenario analysis is conducted by three different scenarios, based on available water resources: the Reference Scenario, Policy Support Scenario, and Water Rights Replacement Scenario.

In 2020, Ordos consumed a total of 1.727 billion cubic meters of water; of this total, 347 million cubic meters was industrial water consumption, an increase of 100 million tons compared to 2015. The "Water Security Plan of Ordos during 14th Five-Year period" projected that in 2025, Ordos is expected to use a total of 1.919 billion cubic meters of water, and the industrial water consumption will increase. Therefore, referring to the increase in industrial water consumption in Ordos during the "13th Five-Year" period, for the Reference Scenario and Policy Support Scenario, it is assumed that 100 million tons of industrial water are added in 2025. Currently, about 70% of the industrial water consumption is used for the coal industry, so the Reference Scenario assumes that 70% of the newly available consumption is used for the coal industry. The Policy Support Scenario assumes that 90% of the newly available consumption is used for the coal industry to promote the development of the coal industry. The Water Rights Replacement Scenario refers to the replacement of water indicators from agriculture, whereby industrial enterprises obtain additional water rights by participating in agricultural water-saving transformation. It is assumed that 200 million tons of industrial water are added in 2025, and 80% is used for the coal industry. Table 4 shows the water resource scenarios.

Table 4

Water resource scenarios (million tons).

Scenarios	Industrial water consumption in 2020	Increment of industrial water in 2025	Proportion of newly available consumption used for coal industry	Water available for coal and related industries in 2025
Reference Scenario	347	100	70%	313
Policy Support Scenario	347	100	90%	333
Water Rights Replacement Scenario	347	200	80%	403

4. Optimization results and analysis

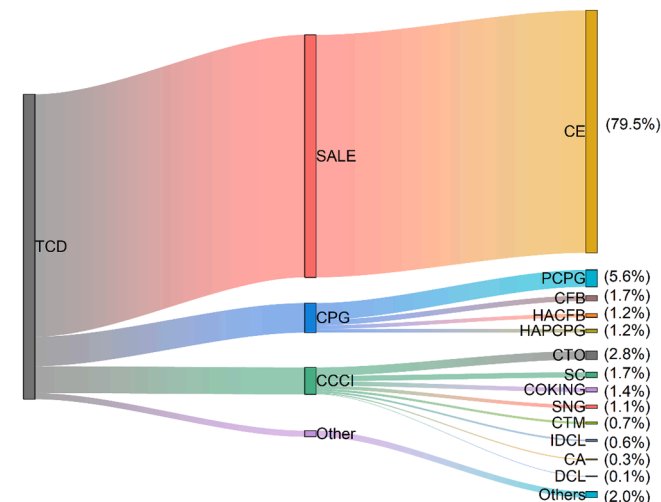
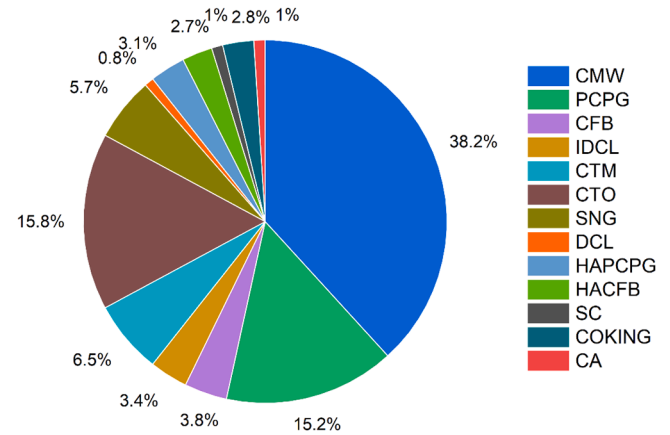
4.1. Reference scenario optimization results and analysis

Fig. 2 shows the coal consumption structure and associated proportion in 2025 in Ordos under the Reference Scenario. The figure shows that the total coal development in Ordos is 747.87 million tons. Of this, approximately 595 million tons, or 79.5%, are exported due to water resource constraints. The local utilization rate of coal (known as the localization rate, which refers to the processing and conversion of power generation, coal chemical and other processes in Ordos) is 20.5 percent, below the Ordos government's target of 40 percent. In the local utilization, the coal technology categories with large coal consumption include PCPG (5.6%), CTO (2.8%) and Others (2.0%), with coal consumption of 41.67 million tons, 20.71 million tons and 15 million tons, respectively. The other categories account for less than 2%.

Fig. 3 shows the proportion of water resource consumption by coal related industries expected in 2025 in Ordos under the Reference Scenario. The figure shows that the total water consumption of coal and related industries in Ordos is expected to reach the maximum limit of 313 million tons; coal washing consumes the most water, at 119.7 million tons, or 38.2%. Power generation consumes 77.77 million tons, and the coal chemical industry consumes 115.56 million tons. The key water-consuming industries include CTO (15.8%), PCPG (15.2%), and CTM (6.5%) processes.

4.2. Optimization results and analysis under different water resource scenarios

The analysis in the section above shows that water resources are an important constraint limiting the development of coal and its related industries. This section reports on the other two modeled scenarios: the Policy Support Scenario and Water Rights Replacement Scenario.

**Fig. 2.** Coal flow diagram under the reference scenario.**Fig. 3.** Proportion of water resources consumption by coal related industries in Ordos under the reference scenario.

4.2.1. Analysis of policy support scenario

Under the Policy Support Scenario, the amount of available water resources increases by 20 million tons compared with the Reference Scenario, and the total amount of coal development remains essentially the same, at 747 million tons. However, the total export of coal decreases slightly, from 594.6 million tons to 588.53 million tons, while the proportion of the coal exports decreases to 78.8%, as shown in Fig. 4. In contrast to this trend, the local utilization rate of coal increases to 21.2 percent. This is due to an increase in overall coal utilization due to more available water. In terms of specific technical capacity, the consumption of indirect oil production and high-aluminum coal for coal-fired power generation increases by 5.957 million tons and 5.761 million tons, respectively; coal consumption decreases by 6.764 million tons, and coal consumption through other technologies remains unchanged. In other words, the production capacity is not expected to change.

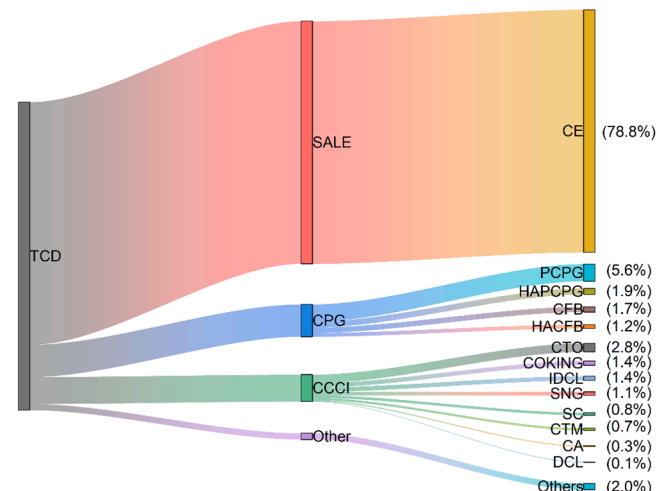
**Fig. 4.** Coal flow diagram under the policy support scenario.

Fig. 5 shows the water resource consumption and its proportion of coal related industries expected in 2025 in Ordos under the Policy Support Scenario. The figures show that the water structure also changes, with the proportion of water used for coal mining and washing dropping to 35.9%, while the water consumption remained almost unchanged at 119.50 million tons. Other important water consuming technologies under this scenario are CTO (14.8%), PCPG (14.3%), and CTM (6.2%) technologies. The water consumption of IDCL production increased by 14.774 million tons, or 7.6%. In addition, the water consumption of HAPCPG increased by 6.572 million tons, while the water consumption of SC decreased by 1.61 million tons. The water consumption of other technologies did not change.

4.2.2. Analysis of Water Rights Replacement Scenario

Fig. 6 shows the coal consumption structure and associated proportion in 2025 in Ordos under the Water Rights Replacement Scenario. Under this scenario, the available water resources increase by 90 million tons compared with the Reference Scenario, and total coal development increases by 20 million tons, reaching 766.85 million tons. The export of coal decreases to 576.26 million tons, accounting for 75.1% of total coal development. The significant increase in available water resources leads the local utilization rate of coal to significantly increase to 24.9%. Production output increases for indirect coal liquefaction, coal-to-methanol, high-alumina pulverized coal for power generation and coking, with the associated coal consumption increasing by 5.957 million tons, 12.383 million tons, 5.761 million tons and 20.567 million tons, respectively. In contrast, coal consumption of semi-coke decreased by 6.764 million tons.

Fig. 7 shows the water resources consumption of coal related industries in the Water Rights Replacement Scenario. The proportion of washing water in coal mining drops to 30.7%. The water consumption of each technology relates to the production capacity, with the coal-to-methanol technology occupying the largest proportion of water consumption, with water consumption of 67.878 million tons, accounting for 17.0%. The coking sector is projected to experience the next largest increase in water consumption, up 16.986 million tons. The total water resource consumption in this scenario is 400.13 million tons, which is lower than the established limit of 403 million tons. This is because the increase in the local utilization rate of coal leads to a corresponding increase in pollutant emissions, with the total amount of SO₂ discharged from coal development and utilization reaching the urban emission reduction target. Therefore, water resources are no longer the primary constraint for coal development and utilization in this scenario.

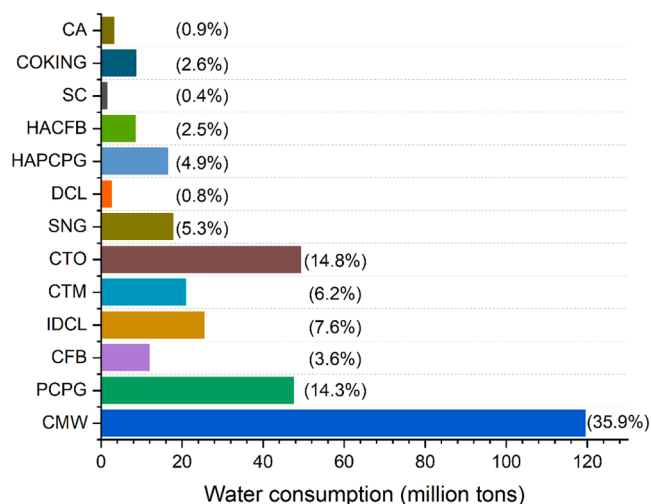


Fig. 5. Water resources consumption by coal related industries in Ordos under the policy support scenario.

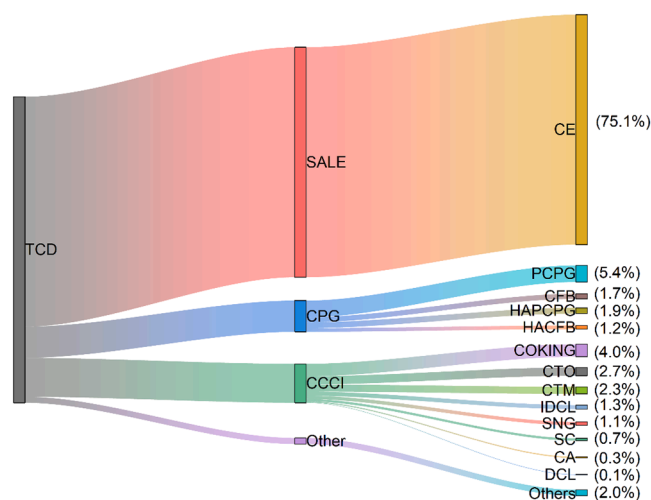


Fig. 6. Coal flow diagram under the Water Rights Replacement Scenario.

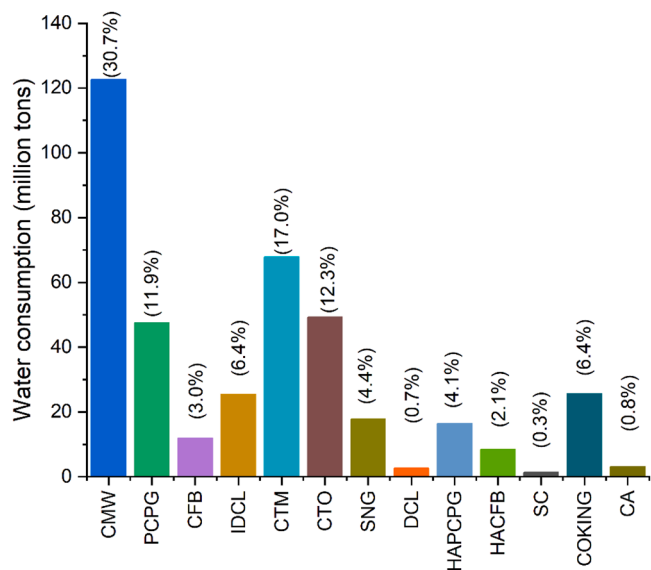


Fig. 7. Water resources consumption by coal related industries in Ordos under the Water Rights Replacement Scenario.

4.2.3. Comparison of the three water resource scenarios

Figs. 8 and 9 present the coal consumption and water consumption, respectively, of coal related industries under different water resource scenarios. The amount of available water resources does not significantly increase in the Policy Support Scenario compared with the Reference Scenario. The structure of coal consumption and water resource consumption does not significantly change; the coal used for power generation increases by 5.761 million tons, and the coal used for the chemical industry slightly decreases.

In contrast, under the Water Rights Replacement Scenario, the amount of available water resources increases by nearly 100 million tons, and the coal consumed by the coal chemical industry and its corresponding water consumption increase significantly, by 31.543 million tons and 77.421 million tons, respectively. The water consumption for IDCL, CTM, HAPCPG, and COKING increased by 14.774 million tons, 47.515 million tons, 6.572 million tons, and 16.986 million tons, respectively. Compared with the Policy Support Scenario, coal consumption and water consumption in coal-fired power generation do not significantly change under the Water Rights Replacement Scenario. The change in available water resources most significantly impacts the coal

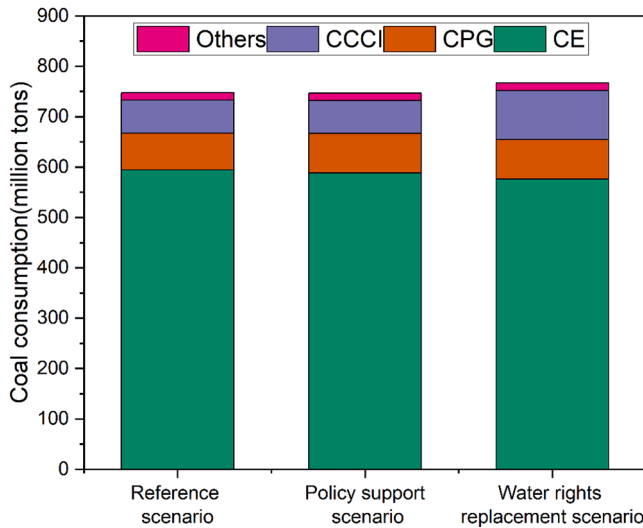


Fig. 8. Coal consumption by coal related industries in Ordos under different scenarios.

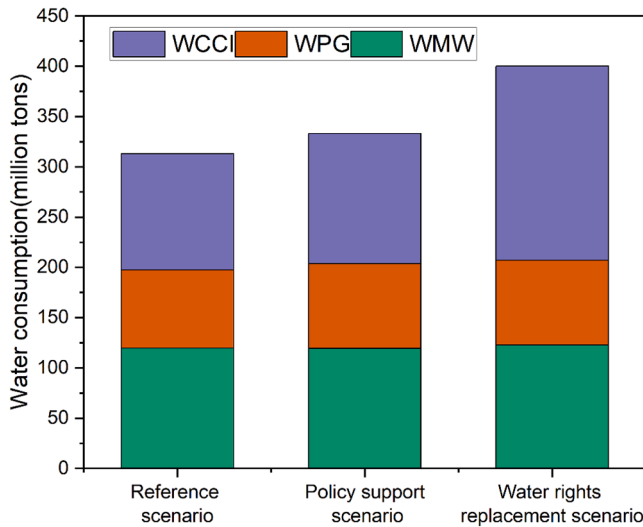


Fig. 9. Water resources consumption by coal related industries in Ordos under different scenarios.

chemical industry.

4.3. Carbon dioxide emission scenario analysis

The continuous growth of carbon dioxide emissions seriously threatens the sustainable development of China's society (Xu and Xu, 2023; HAN et al., 2023). Actively and prudently achieving a carbon peak and carbon neutrality is key to China's national strategy for addressing climate change. Achieving these goals requires all regions to promote high-quality development. Ordos has formed an industrial structure that includes coal mining, coal power, and the coal chemical industry, based on coal energy consumption. In 2020, coal consumption accounted for 91.3% of the total primary energy consumption. However, coal related industries are a significant source of carbon dioxide emissions. This section analyzes how a carbon dioxide emission constraint impacts the coal industry.

Here, the basic scenario assesses the carbon dioxide emissions corresponding to the total energy consumption of Ordos in 2025; the impact of carbon dioxide emission restrictions on the coal industry is then assessed when the carbon dioxide emissions are reduced by 15%, 20%,

and 25%. Fig. 10 shows the results. In the basic scenario, carbon emission is total of 226.81 million tons, and does not create a constraint, because water resources are the larger constraint. When the carbon dioxide emissions are reduced by 15%, 20%, and 25%, the carbon dioxide emissions are 223.15 million tons, 210.02 million tons, and 196.9 million tons, respectively. The output from technical categories is gradually affected by the carbon emission limits.

Fig. 11 shows that the increased emission limits do not change the output of some technology categories; some increase and some decrease. Productions from CTM, SC, and COKING increase gradually, while IDCL production and CTO production decrease. In particular, when carbon emissions are reduced by 25%, the coal consumption for IDCL production and CTO production is reduced by 7.176 million tons and 14.494 million tons, respectively. Thus, the largest impact from the carbon emissions reduction scenario is seen for IDCL production and CTO production. In other words, to reduce carbon emissions, IDCL and CTO should be reduced.

5. Conclusions and policy implications

5.1. Conclusions

According to the concepts of technology selection based on coal type and capacity determination based on constraints, this paper proposes a modeling method that combines superstructure modeling and MIP mathematical method. The production capacity allocation of the coal industry expected in 2025 in Ordos, China is analyzed based on water resource, pollutant emission, and carbon dioxide emission constraints.

Three scenarios are set for analysis: a Reference Scenario, Policy Support Scenario, and Water Rights Replacement Scenario. The optimization results of coal industry under the water resource constraint show that an increase in the available water resources is associated with an increase in the local utilization rate of coal. Compared with the local coal utilization rate of 20.5% under the Reference Scenario, the value under the Water Rights Replacement Scenario reaches 24.9%. However, this value does not continue to increase, due to pollutant emission limitations. This scenario highlights an effective pathway to improve the local utilization rate of coal to reduce the pollutant emission per ton of product through technological innovation. Increases in available water resources impact the coal chemical industry most significantly. Under the Water Rights Replacement Scenario, coal consumption and water consumption by the coal chemical industry increased by 31.543 million tons (47.89%) and 77.421 million tons (67.08%), respectively.

Moreover, the impact of carbon dioxide emission limitation on the capacity allocation of the coal industry is demonstrated by comparing

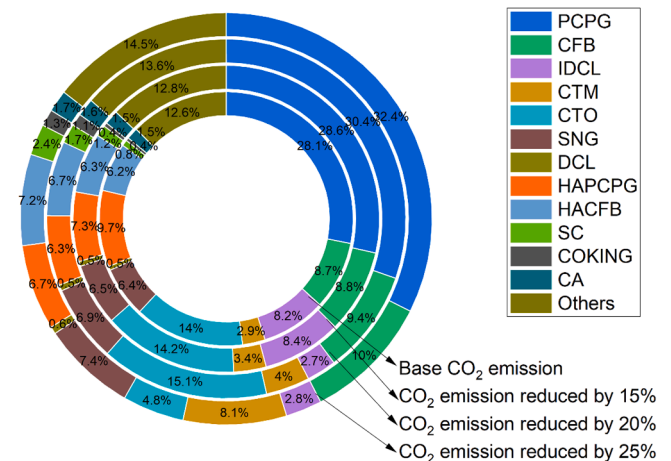


Fig. 10. Proportion of carbon dioxide emissions under different carbon emission reduction scenarios.

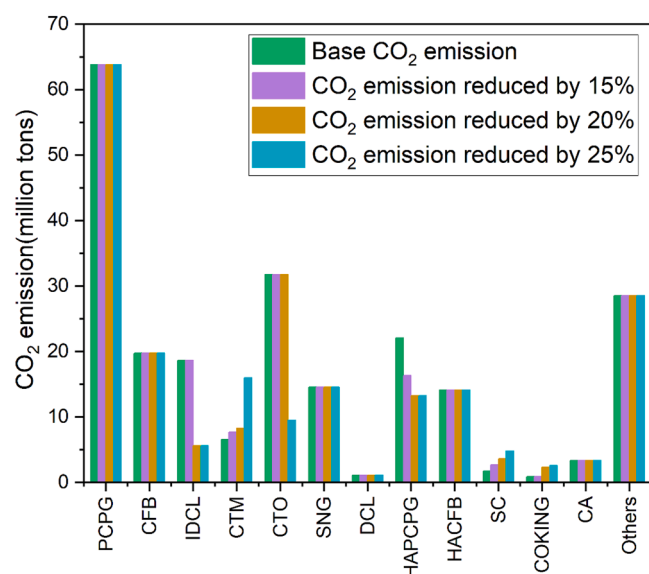


Fig. 11. Carbon dioxide emissions under different carbon emission reduction scenarios.

the scenarios. The results indicate that the production of IDCL and CTO should be reduced when the carbon dioxide emission level is expected to be significantly reduced. When the modeled carbon dioxide emissions are reduced by 25%, the coal consumption from IDCL and CTO is reduced by 7.176 million tons and 14.494 million tons, respectively, both of which reflect a 70% reduction.

5.2. Policy implications

1) Adhere to the priority strategy of energy conservation and water conservation. On the one hand, advocate energy and resources conservation and guide industrial enterprises to conserve water consumption. On the other hand, strengthen the recycling of water resources, and improve the reuse rate of mine water, sewage discharge and coal chemical wastewater, to achieve the efficient recycling of water for the coal industry.

2) Taking into account the limited resources and environmental carrying capacity, reducing water consumption, pollutants and carbon dioxide emissions per unit product can effectively improve the local coal utilization rate. The government can guide enterprises, universities, and research institutes to establish community of sci-tech innovation, focus on the pain points of high energy consumption and large emissions in the coal industry, and work together to reduce water consumption, pollutants and carbon dioxide emissions per unit product of coal utilization technology, play the innovation coordinated effect of industry, university, research and application, and improve the efficiency of technological innovation in the coal industry.

3) Anchoring the goals of "Dual Carbon", China is in the transition stage from "dual control" of energy consumption to "dual control" of total and intensity of carbon emissions. The coal industry is the main source of carbon dioxide emission, and the key industry of carbon dioxide emission reduction. The local and central governments should comprehensively consider the relationship between carbon dioxide emission and the secure supply of coal-fired power and coal chemical products, deploy the coal industry scientifically and rationally, and promote high-quality economic development.

The model in this paper focuses on how to address regional coal industry capacity allocation considering multiple constraints. The change in technical parameters over time and the optimal allocation in a larger area were not considered in this paper. To address this, future research could be done to extend to a larger geographic area and

incorporate more industrial fields. Besides, the planning method proposed in this study is applicable to investigate the optimized results of the coal industry under given constrained water resources and obtain the policy implications on the coal industry for other countries or regions accordingly. However, it needs a concrete analysis based on the method for specific situations when it comes to other counties of regions because of the difference in the current situation of the coal resources and the coal industry as well as the constraints of available water resources and carbon dioxide emission.

CRedit authorship contribution statement

Chao Cheng: Conceptualization, Methodology, Validation, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Dan Gao:** Conceptualization, Methodology, Writing – review & editing, Supervision, Resources. **Heng Zhang:** Data curation, Investigation, Writing – review & editing. **Zipeng Xu:** Writing – original draft, Software, Data curation, Formal analysis. **Jiguang Huang:** Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

Acknowledgments

This work was supported by the Fundamental Research Funds for the Central Universities (2023MS094) and the Joint Strategic Research and Consulting Project of Chinese Academy of Engineering and National Natural Science Foundation of China (2022-ZCQ-07).

References

- Biegler, L.T., Grossmann, I.E., Westerberg, A.W., 1997. Systematic Methods of Chemical Process Design. Prentice Hall, New Jersey.
- Chen, S., Song, Y., Zhang, M., 2021. Study on the sustainability evaluation and development path selection of China's coal base from the perspective of spatial field. *Energy* 215, 119143.
- Chinese Academy Of Engineering, 2017. Strategic Study on Major Issues of Sustainable Development and Regional Coordinated Development of China's Modern Coal Chemical Industry. Chinese Academy Of Engineering, Beijing.
- Cui, Z., Zhang, J., Wu, D., Cai, X., Wang, H., Zhang, W., Chen, J., 2020. Hybrid many-objective particle swarm optimization algorithm for green coal production problem. *Inf. Sci.* 518, 256–271.
- Gomes, C.M., Kneipp, J.M., Kruglianskas, I., Da Rosa, L.A.B., Bichueti, R.S., 2014. Management for sustainability in companies of the mining sector: an analysis of the main factors related with the business performance. *J. Clean. Prod.* 84, 84–93.
- Guo, M., Xu, Y., 2018. Coal-to-liquids projects in China under water and carbon constraints. *Energy Policy* 117, 58–65.
- Han, X., Li, Y., Nie, L., Huang, X., Deng, Y., Yan, J., Kourkoumpas, D., Karellas, S., 2023. Comparative life cycle greenhouse gas emissions assessment of battery energy storage technologies for grid applications. *J. Clean. Prod.* 392, 136251.
- Hassan, S.T., Batool, B., Sadiq, M., Zhu, B., 2022. How do green energy investment, economic policy uncertainty, and natural resources affect greenhouse gas emissions? A Markov-switching equilibrium approach. *Environ. Impact Assess.* 97, 106887.
- Hou, Y., Long, R., Chen, H., Zhang, L., 2018. Research on the sustainable development of China's coal cities based on lock-in effect. *Resour. Policy* 59, 479–486.
- Huang, W., Ma, D., Chen, W., 2017. Connecting water and energy: assessing the impacts of carbon and water constraints on China's power sector. *Appl. Energy* 185, 1497–1505.
- Huang, Y., Yi, Q., Kang, J., Wu, G., Feng, J., Xie, K., Li, W., 2018. A nonlinear programming approach to strategic planning of coal chemical industry with carbon constraints in China. *Energy Procedia* 152, 1236–1241.

- Huang, Y., Yi, Q., Kang, J., Zhang, Y., Li, W., Feng, J., Xie, K., 2019. Investigation and optimization analysis on deployment of China coal chemical industry under carbon emission constraints. *Appl. Energy* 254, 113684.
- Ji, L., Zhang, B., Huang, G., Xie, Y., Niu, D., 2018. GHG-mitigation oriented and coal-consumption constrained inexact robust model for regional energy structure adjustment—a case study for Jiangsu Province, China. *Renew. Energy* 123, 549–562.
- Jiang, T., Yu, Y., Jahanger, A., Balsalobre-Lorente, D., 2022. Structural emissions reduction of China's power and heating industry under the goal of "double carbon": a perspective from input-output analysis. *Sustain. Prod. Consum.* 31, 346–356.
- Jie, D., Xu, X., Guo, F., 2021. The future of coal supply in China based on non-fossil energy development and carbon price strategies. *Energy* 220, 119644.
- Kong, H., Li, Z., Yu, Z., Zhang, J., Wang, H., Wang, J., Gao, D., 2021. Environmental and economic multi-objective optimization of comprehensive energy industry: a case study. *Energy* 237, 121534.
- Kong, Z., Dong, X., Jiang, Q., 2019. Forecasting the development of China's coal-to-liquid industry under security, economic and environmental constraints. *Energy Econ.* 80, 253–266.
- Leao, A.A.S., Toledo, F.M.B., Oliveira, J.F., Carravilla, M.A., Alvarez-Valdés, R., 2020. Irregular packing problems: a review of mathematical models. *Eur. J. Oper. Res.* 282 (3), 803–822.
- Li, J., Hu, S., 2017. History and future of the coal and coal chemical industry in China. *Resour. Conserv. Recycl.* 124, 13–24.
- Li, J., Zhang, Y., Deng, Y., Xu, D., Tian, Y., Xie, K., 2021. Water consumption and conservation assessment of the coal power industry in China. *Sustain. Energy Technol. Assess.* 47, 101464.
- Li, Z., Yang, F., Huang, B., 2023. The assessment of energy-related greenhouse gas emissions in China's chemical industry. *Sustain. Prod. Consum.* 37, 112–125.
- Liu, H., Ren, Y., Wang, N., 2021. Energy efficiency rebound effect research of China's coal industry. *Energy Rep.* 7, 5475–5482.
- Liu, L., Cui, G., Chen, J., Huang, X., Li, D., 2022. Two-stage superstructure model for optimization of distributed energy systems (DES) part I: model development and verification. *Energy* 245, 123227.
- Liu, Y., Bian, J., Li, X., Liu, S., Lageson, D., Yin, Y., 2020. The optimization of regional industrial structure under the water-energy constraint: a case study on Hebei Province in China. *Energy Policy* 143, 111558.
- Liu, Y., Zhang, Z., Huang, X., 2023. A novel coupling cloud decision-making framework for investigating population-economy-water consumption nexus in the Inner Mongolia reaches of the Yellow River, China. *Environ. Impact Assess.* 98, 106923.
- Long, R., Ren, Y., Wu, M., 2022. Differential decomposition of total-factor energy efficiency in Chinese coal mining cities considering environmental constraints: a dynamic and static perspective. *Resour. Policy* 79, 102993.
- Misrol, M.A., Wan Alwi, S.R., Lim, J.S., Abd Manan, Z., 2021. Optimization of energy-water-waste nexus at district level: a techno-economic approach. *Renew. Sustain. Energy Rev.* 152, 111637.
- Ordos Municipal People's Government, 2017. Action plan for improving the comprehensive competitiveness of coal industry in Ordos (2016–2018).
- Ordos Municipal People's Government, 2022. Comprehensive energy development plan of Ordos during the "fourteenth five year plan".
- Shi, J., Huang, W., Han, H., Xu, C., 2021. Pollution control of wastewater from the coal chemical industry in China: environmental management policy and technical standards. *Renew. Sustain. Energy Rev.* 143, 110883.
- Suopajarvi, L., Poelzer, G.A., Ejdemo, T., Klyuchnikova, E., Korchak, E., Nygaard, V., 2016. Social sustainability in northern mining communities: a study of the European North and Northwest Russia. *Resour. Policy* 47, 61–68.
- Wang, C., Olsson, G., Liu, Y., 2018. Coal-fired power industry water-energy-emission nexus: a multi-objective optimization. *J. Clean. Prod.* 203, 367–375.
- Xie, K., 2014. Clean and Efficient Conversion of Coal. Science Press, Beijing.
- Xiong, R., Wei, C., 2017. Current status and technology trends of zero liquid discharge at coal chemical industry in China. *J. Water Process Eng.* 19, 346–351.
- Xu, B., Xu, R., 2023. Assessing the carbon intensity of the heavy industry in China: using a nonparametric econometric model. *Environ. Impact Assess.* 98, 106925.
- Xu, M., Li, C., Wang, X., Cai, Y., Yue, W., 2018. Optimal water utilization and allocation in industrial sectors based on water footprint accounting in Dalian City, China. *J. Clean. Prod.* 176, 1283–1291.
- Xu, X., Xu, X., 2021. Can resource policy adjustments effectively curb regional "resource curse"? New evidences from the "energy golden triangle area" of China. *Resour. Policy* 73, 102146.
- Yang, J., Wu, J., He, T., Li, L., Han, D., Wang, Z., Wu, J., 2016. Energy gases and related carbon emissions in China. *Resour. Conserv. Recycl.* 113.
- Zeng, L., Guo, J., Wang, B., Lv, J., Wang, Q., 2019. Analyzing sustainability of Chinese coal cities using a decision tree modeling approach. *Resour. Policy* 64, 101501.
- Zhang, R., Qie, X., Hu, Y., Chen, X., 2022. Does de-capacity policy promote the efficient and green development of the coal industry? –Based on the evidence of China. *Resour. Policy* 77, 102717.
- Zhang, Y., Yuan, Z., Margni, M., Bulle, C., Hua, H., Jiang, S., Liu, X., 2019. Intensive carbon dioxide emission of coal chemical industry in China. *Appl. Energy* 236, 540–550.
- Zhao, J., Zhou, L., Zhou, W., Ren, H., Yu, Y., Wang, F., Ma, T., 2021a. Techno-economic analysis and comparison of coal-based chemical technologies with consideration of water resources scarcity. *Energy Strategy Rev.* 38, 100754.
- Zhao, L., Liu, Z., Cheng, L., 2021b. How will China's coal industry develop in the future? A quantitative analysis with policy implications. *Energy* 235, 121406.
- Zhao, Y., Yang, Y., Leszek, S., Wang, X., 2021c. Experience in the transformation process of "coal city" to "beautiful city": taking Jiaozuo City as an example. *Energy Policy* 150, 112164.
- Zuo, Z., Guo, H., Li, Y., Cheng, J., 2022. A two-stage DEA evaluation of Chinese mining industry technological innovation efficiency and eco-efficiency. *Environ. Impact Assess.* 94, 106762.